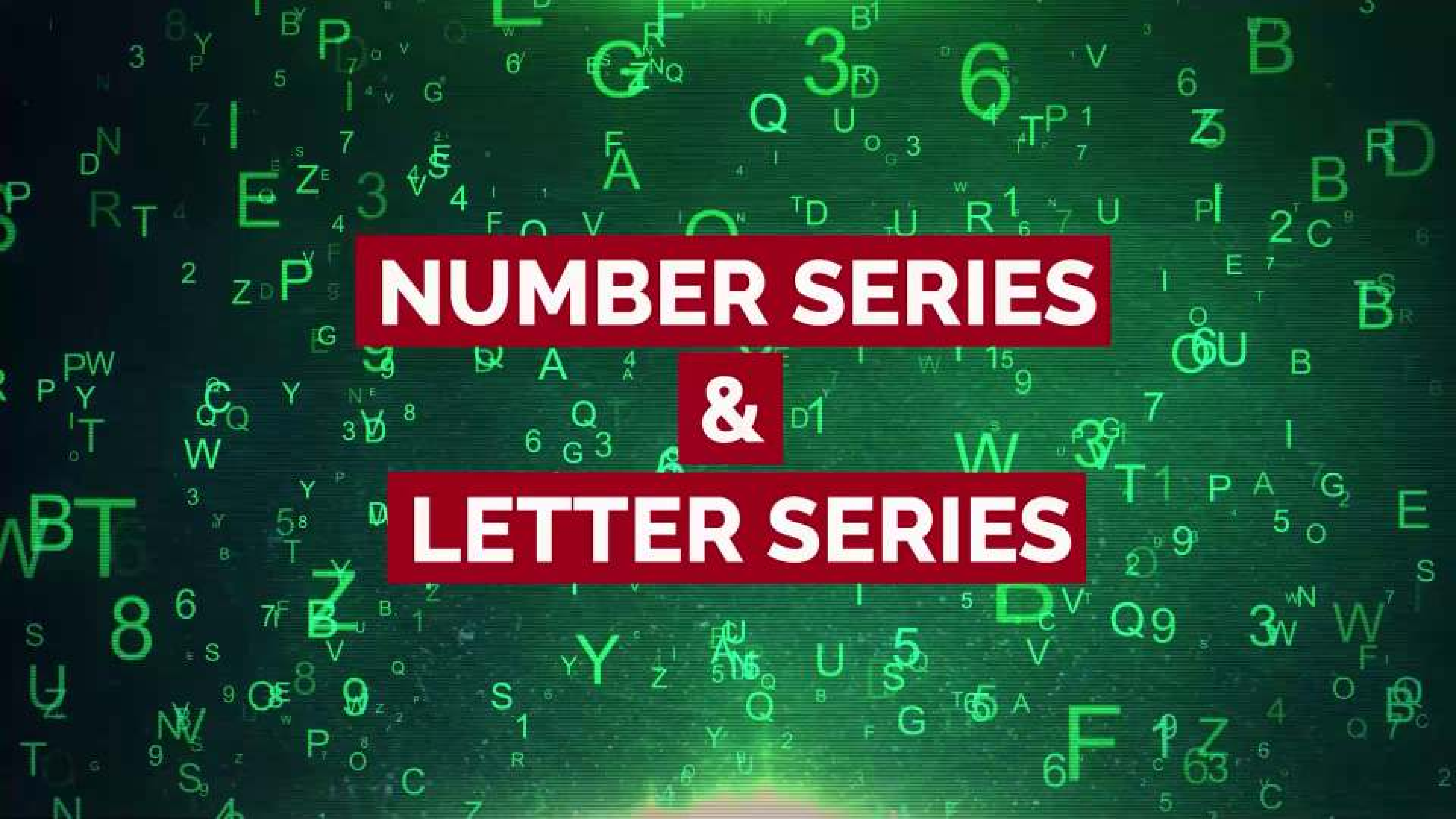


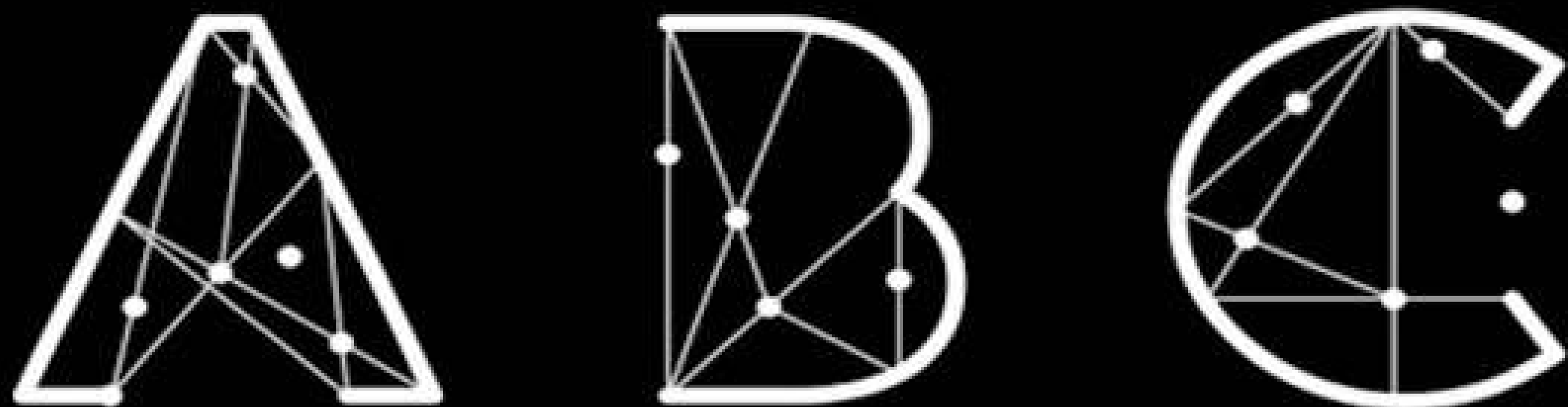


NUMBER SERIES

&

LETTER SERIES

NUMBER SERIES - LETTER SERIES



- by YASH JAIN

Solve This Number Puzzle

Find the Next Number in this series:

5, 7, 12, 19, 31, 50, ?



back To basics



Some Important Points

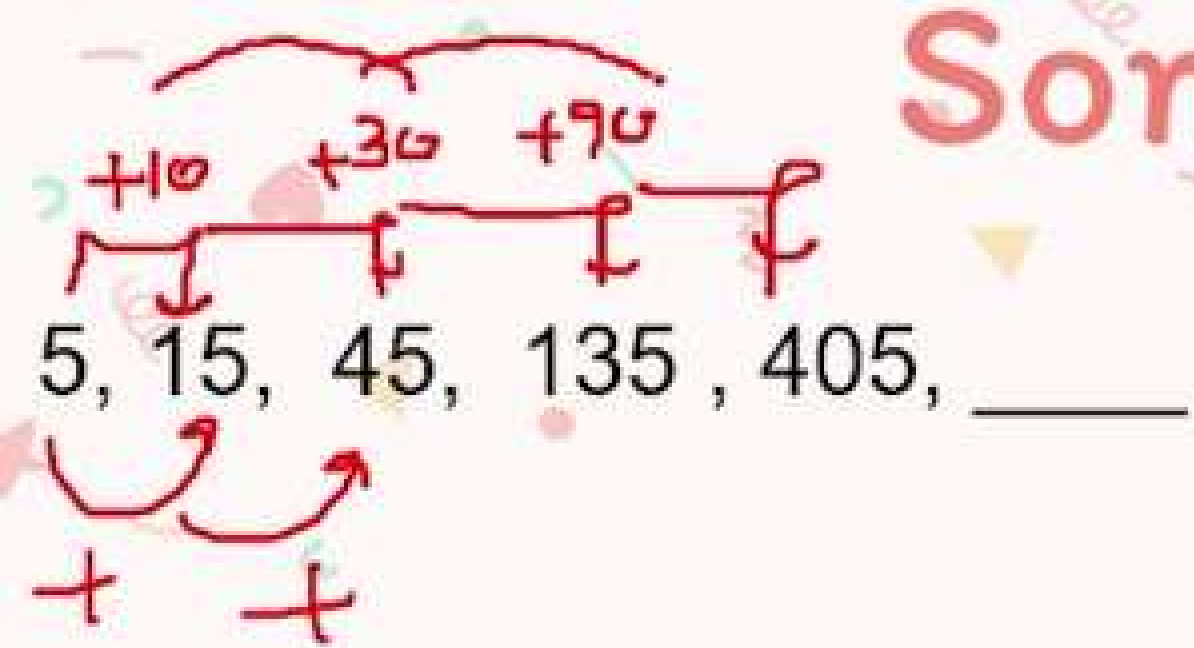
5, 15, 45, 135, 405, _____

524288 65536 8192 1024 128 16 _____



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Some Important Points



addition
multiplication
Mix (A+m)

524288 65536 8192 1024 128 16 ____

Some Important Points

$(15 \times 2) + 15$
 $(45 \times 2) + 45$
5, 15, 45, 135, 405, ____
5 $\longrightarrow$ 15
 $\rightarrow (5 \times 2) + 5$

addition
multiplication
Mix(A+m)

524288 65536 8192 1024 128 16 ____

Some Important Points

5, 15, 45, 135, 405, _____
→ inc ↑↑↑

add x
mul ✓
x (mul + add) ✓

524288 65536 8192 1024 128 16 _____

Some Important Points

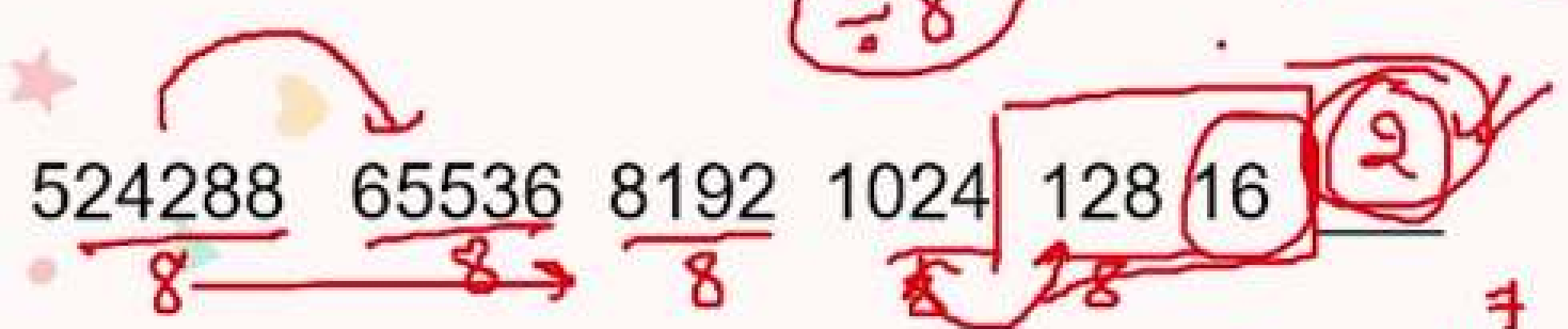


$$\begin{array}{r} 405 \\ \times 3 \\ \hline \end{array}$$

1215

$$\div 8$$

$$16/8 = 2$$



Sub. X
div. ✓
div + sub. ✓

$$\begin{array}{r} 524288 \\ \div 8 \\ \hline 65536 \end{array}$$

$$\begin{array}{r} 1024 \\ \div 128 \\ \hline 8 \end{array}$$

$$\frac{128}{16} = \frac{2^7}{2^4} = 2^{7-4} = 2^3 = 8$$

$$\frac{2^{10}}{2^4} = 2^6 = 8 \quad \frac{a^m}{a^n} = a^{m-n}$$

Some Important Points

961 1024 1089 — 1225



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Some Important Points



Squares 1 to 50

$1^2 = 1$	$11^2 = 121$	$21^2 = 441$	$31^2 = 961$	$41^2 = 1681$
$2^2 = 4$	$12^2 = 144$	$22^2 = 484$	$32^2 = 1024$	$42^2 = 1764$
$3^2 = 9$	$13^2 = 169$	$23^2 = 529$	$33^2 = 1089$	$43^2 = 1849$
$4^2 = 16$	$14^2 = 196$	$24^2 = 576$	$34^2 = 1156$	$44^2 = 1936$
$5^2 = 25$	$15^2 = 225$	$25^2 = 625$	$35^2 = 1225$	$45^2 = 2025$
$6^2 = 36$	$16^2 = 256$	$26^2 = 676$	$36^2 = 1296$	$46^2 = 2116$
$7^2 = 49$	$17^2 = 289$	$27^2 = 729$	$37^2 = 1369$	$47^2 = 2209$
$8^2 = 64$	$18^2 = 324$	$28^2 = 784$	$38^2 = 1444$	$48^2 = 2304$
$9^2 = 81$	$19^2 = 361$	$29^2 = 841$	$39^2 = 1521$	$49^2 = 2401$
$10^2 = 100$	$20^2 = 400$	$30^2 = 900$	$40^2 = 1600$	$50^2 = 2500$

$$\begin{array}{ccccc} \underline{961} & \underline{1024} & \underline{1089} & \underline{1156} & \underline{1225} \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 31^2 & 32^2 & 33^2 & 34^2 & 35^2 \end{array}$$



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Some Important Points



Cubes 1 to 20

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = \underline{729}$$

$$10^3 = \underline{1000}$$

$$11^3 = 1331$$

$$12^3 = \underline{1728}$$

$$13^3 = \underline{2197}$$

$$14^3 = 2744$$

$$15^3 = 3375$$

$$16^3 = 4096$$

$$17^3 = 4913$$

$$18^3 = 5832$$

$$19^3 = 6859$$

$$20^3 = 8000$$

✓✓ 729 1000 1331 1728 2197

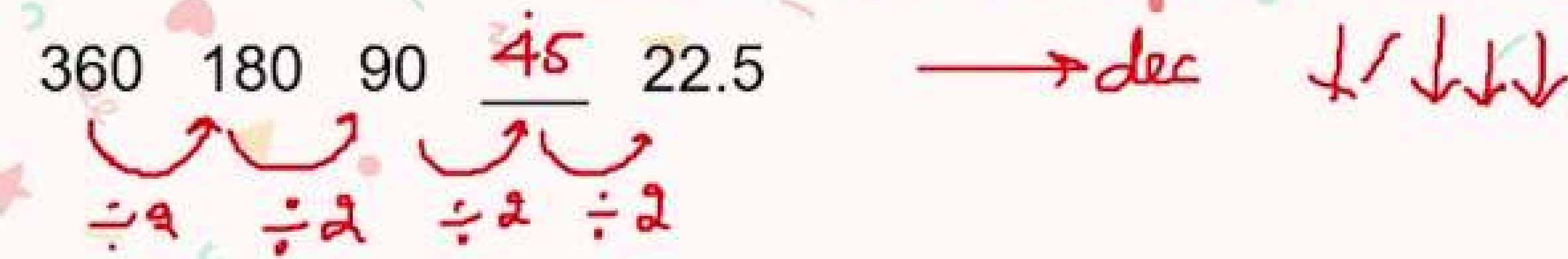
$9^3 = 729$
 $10^3 = 1000$
 $11^3 = 1331$
 $12^3 = 1728$
 $13^3 = 2197$

Arrows indicate the sequence of cubes from 9 to 13.

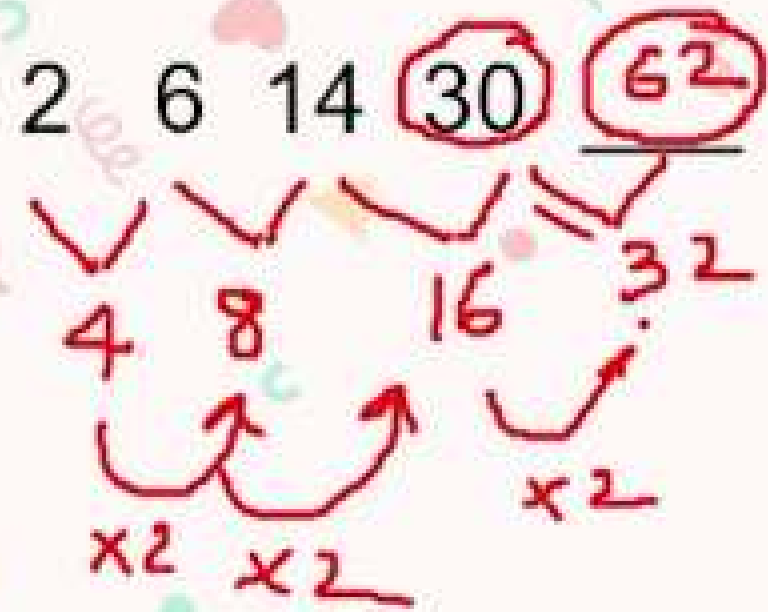


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Some Important Points



Some Important Points

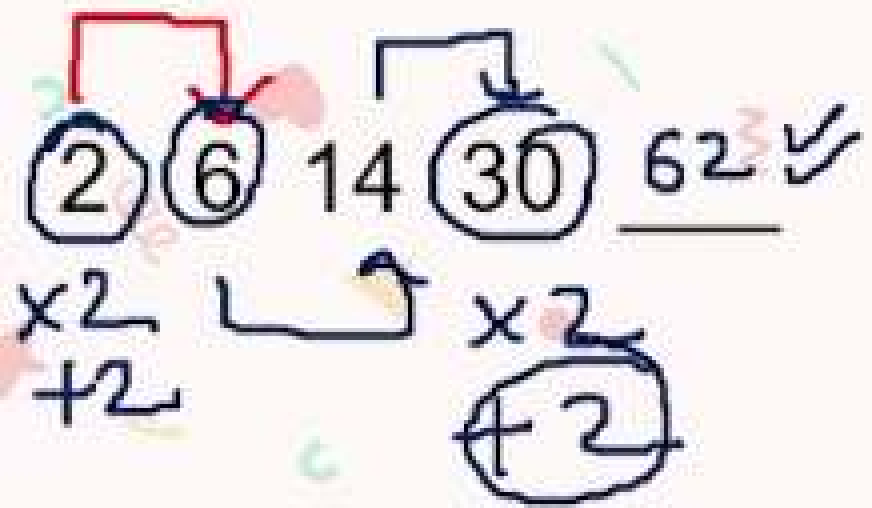


Some Important Points

2 6 14 30 —

Brute Force

Some Important Points



$$14 \longrightarrow 30$$

$$\hookrightarrow (14 \times p) + m$$

$$(14 \times 2) + 2$$

$$\Rightarrow 30 \checkmark$$

$$2 \longrightarrow 6$$

$$\hookrightarrow (2 \otimes p) \oplus m$$

$$(2 \times 2) + 2 \Rightarrow 6$$

$$6 \longrightarrow 14$$

$$(6 \times p) + m$$

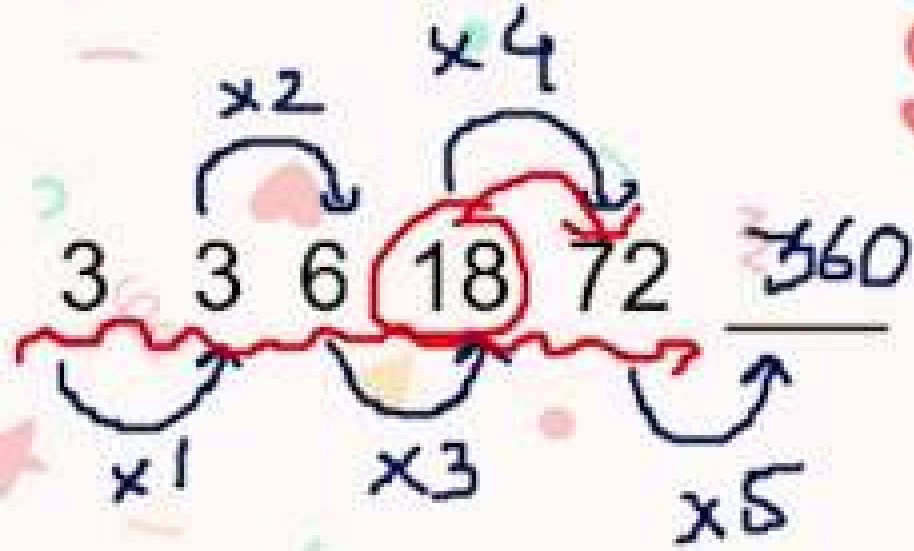
$$(6 \times 2) + 2 = 14 \checkmark$$

$$p = 2$$

$$m = 2$$

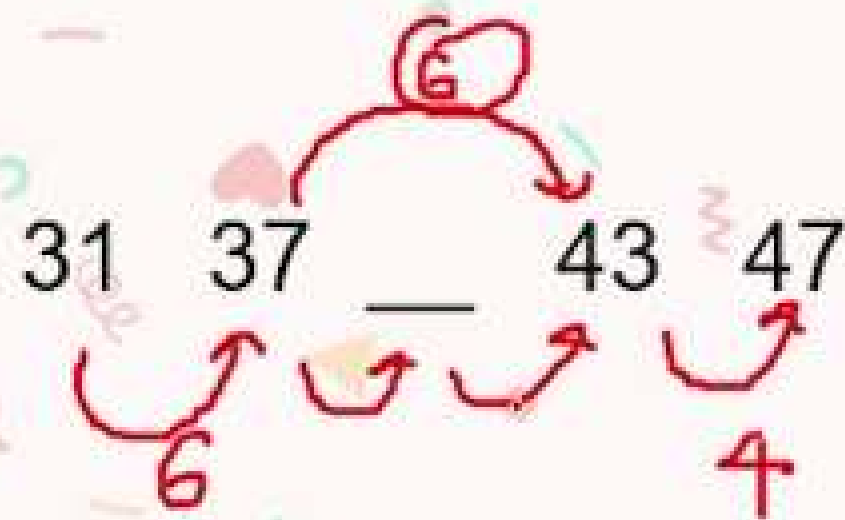
$$\underline{\underline{\times 2 + 2}}$$

Some Important Points



$$72 \times 5 = \underline{\underline{360}}$$

Some Important Points



31 37 41 43 47

Prime numbers between 1 and ~~200~~

2	3	5	7	11	13	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

Some Important Points

↓ ↓ ↓
6 15 35 77 143

$$2 \times 3 = 6 \quad \checkmark$$

$$3 \times 5 = 15 \quad \checkmark$$

$$5 \times 7 = 35 \quad \checkmark$$

$$7 \times 11 = \textcircled{77} \quad \checkmark$$

$$11 \times 13 = 143 \quad \checkmark$$

Prime numbers between 1 and 200

<u>2</u>	<u>3</u>	<u>5</u>	<u>7</u>	<u>11</u>	<u>13</u>	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

Some Important Points

21 24 29 36 47 60
3 5 7 11 13

$47 + 13 = \underline{60}$ Prime numbers between 1 and 200

2	3	5	7	11	13	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

3 5 7 9 11 simple

3 5 7 11 13

Miscellaneous Series

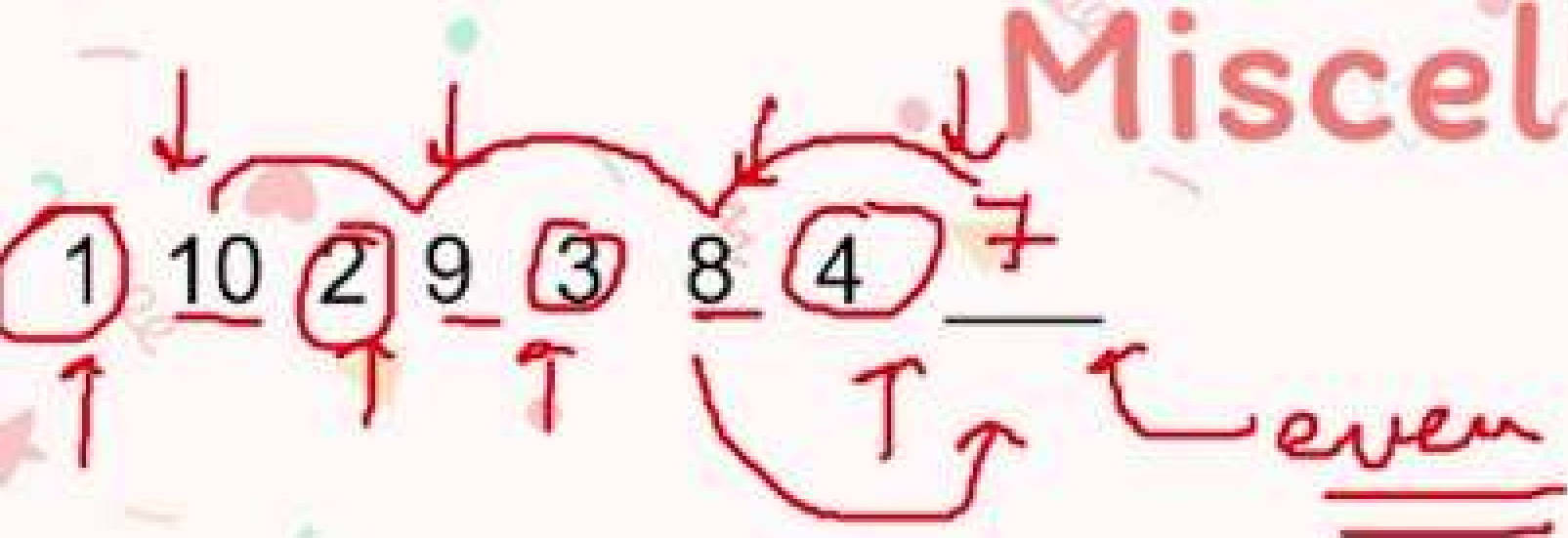
1 10 2 9 3 8 4 —

The diagram shows a sequence of numbers: 1, 10, 2, 9, 3, 8, 4, followed by a blank line. Red curved arrows connect the numbers in a zig-zag pattern: from 1 to 10, 10 to 2, 2 to 9, 9 to 3, 3 to 8, 8 to 4. This illustrates a pattern where the numbers alternate between increasing by 9 and decreasing by 9.

breaking

alternating

Miscellaneous Series



Miscellaneous Series

2 4 6 10 16 26 42 ____

fibonacci series

prev (2) sum = third

Miscellaneous Series

2 4 6 10 16 26 42 68 ✓
↑ ↑
↓ ↓
↓ ↓

$$2 + 4 = 6$$

$$4 + 6 = 10$$

$$6 + 10 = 16$$

$$10 + 16 = 26$$

$$16 + 26 = 42$$

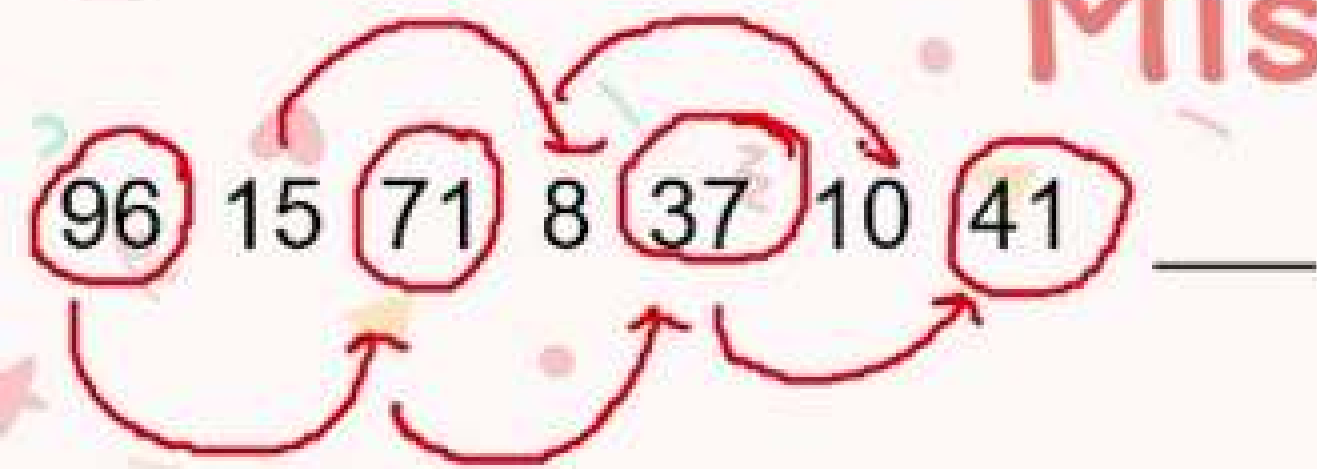
$$26 + 42 = 68 ✓$$

Miscellaneous Series

96 15 71 8 37 10 41 —

breakdown

Miscellaneous Series



Miscellaneous Series

